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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY!



Facial Nerve Palsy

By: Dr. Demerew Dejene (Assistant Professor of Oral and Maxillofacial Surgery)

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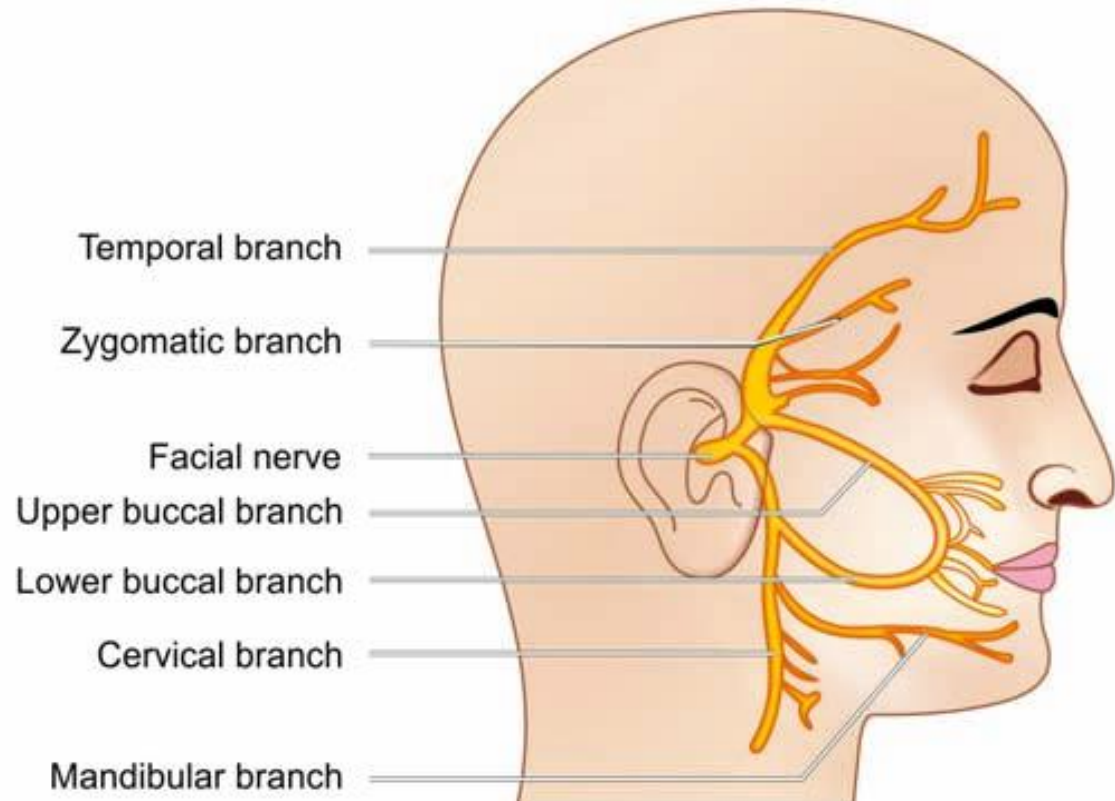
Outline

- ❑ Introduction
- ❑ Etiology
- ❑ Assessment
- ❑ Management
- ❑ Complications
- ❑ References

Introduction

- ❑ The face and perioral regions are among the body regions with highest density of peripheral nerve receptors.
- ❑ The facial nerve?

Facial nerve anatomy



VII

- ✓ It supplies the motor innervation to muscles of facial expressions and also for elevation of the hyoid bone.
- ✓ The motor root supplies to the muscles of the face, scalp, and auricle, the buccinator, platysma, stapedius, stylohyoid and posterior belly of digastric muscle.

VII

- Believed to transmit the preganglionic parasympathetic(secretomotor)fibers to the submandibular and sublingual salivary glands, the lacrimal gland and gland soft palate ,the nose, and the pharynx.

VII

- ▣ The sensory root conveys from the chorda tympani nerve, the fibers of taste for the anterior two third of the tongue and from the palatine and greater petrosal nerves, the fibers of taste from the soft palate

Damage

- ❑ Paresis or paralysis of the facial muscles is **functionally** and **cosmetically** cataclysmic.
 - Loss of motor control of the lips and cheeks makes drinking and mastication difficult
 - Drooling of saliva out side the mouth
 - pain, infection and corneal ulceration
 - Facial expression becomes asymmetric and
 - Socially embarrassing

Etiology

- ❑ Various etiological factors for facial paralysis
- ❑ Broadly classified into three major groups
 - Intracranial (central cause)
 - Intratemporal
 - Extracranial

Etiology

- ❑ Intracranial (central cause)
 - Vascular abnormalities
 - CNS degenerative diseases
 - Tumors of the intracranial cavity
 - Trauma to the brain
 - Congenital abnormalities and agenesis

Etiology

❑ Intratemporal

- Bacterial and viral infection
- Cholesteatoma
- Blunt temporal bone trauma
 - ❑ Longitudinal and horizontal fractures of the temporal bone
 - ❑ Gunshot wounds
- Tumors invading the middle ear, mastoid and facial nerve
- Iatrogenic causes

Etiology

❑ Extracranial

- Malignant tumors of the parotid gland
- Trauma—lacerations, gunshot wounds
- Iatrogenic causes
 - ❑ Orthognathic surgery
 - ❑ TMJ arthrotomy, arthroscopy
 - ❑ Parotid gland surgery
 - ❑ Facial esthetic surgery
- Primary tumors of the facial nerve
- Malignant tumors of the ascending ramus of the mandible, pterygoid region and skin

Rainer Schmelzeisen et al. reported following causes of facial nerve palsy in 1999:

<i>Congenital</i>	congenital nuclear aplasia- rare Congenital cholesteatoma MelkerssonRosenthal syndrome + lingua plicata
<i>Neurologic</i>	Myasthenia gravis Multiple sclerosis GuillainBarré syndrome
<i>Neoplastic</i>	Facial nerve tumors Glomus tumors Meningiomas, acaustic neuroma Parotid tumors Temporal bone/
<i>Infectious</i>	Otitis media, mastoiditis Bacterial causes (diphtheria, tuberculosis) Viral causes
<i>Iatrogenic</i>	Parotidectomy Rhytidectomy Lateral skull base surgery
<i>Traumatic</i>	Temporal bone fractures Penetrating trauma (gunshot) Facial lacerations High altitude palsy
<i>Other cause</i>	Toxic Metabolic Idiopathic

Evaluation of Facial Nerve Functions

□ History

- Acute onset on awakening - typical of Bell's palsy
- Sudden onset may be also of infectious or inflammatory
- History of infection or rashes
- Patients with neoplasm progress paresis
- In trauma definitive history given by patient
- Full otologic history is obtained

Evaluation of Facial Nerve Functions

- ❑ Examination of face at rest and in motion
 - The symmetry of the forehead wrinkling - patient is asked to raise the brow
 - Eye closure - orbicularis oculi muscle and absence of visible upward rotation and exposure of sclera.
 - Whistling
 - Blowing - allows the detection of asymmetries of the perioral muscles
 - Pure taste sensation is carried out using samples of sweat, bitter, acid and salty substances on the anterior tongue.

Evaluation of Facial Nerve Functions

- Imaging /tests
 - CT scan for skull base fracture
 - MRI to detect intracranial lesions
 - Electro diagnostic tests
 - EMG
 - ENog
 - NET

Classification of facial palsy

Grade	Characteristics
I. Normal function	Normal facial function in all areas
II. Mild dysfunction	<u>Slight weakness</u> , Slight synkinesis F-moderate to good function E-complete closure with minimal effort M-slight asymmetry
III. Moderate dysfunction	<u>Obvious weakness</u> , but not disfiguring asymmetry F – slight to moderate movement E – complete closure with effort M –slight weak with maximum effort
IV. Moderately Severe dysfunction	Obvious weakness with disfiguring asymmetry F- none E- Incomplete closure M- Asymmetrical with maximum effort
V. Severe dysfunction	At rest asymmetry F-none E-incomplete closure M-slight movement
VI. Total paralysis	No movement

House-Brackmann (1985)

Classification

- ❑ Facial nerve injuries are also classified according to **Seddon** or **Sunderland**?
- ❑ Neurapraxia
 - mildest form of nerve injury and the nerve remains intact
 - Mild, temporary injury caused by compression or retraction
 - Temporary blockage of nerve conduction
 - resolves with return of facial mobility within **few weeks**

Classification

□ Axonotmesis

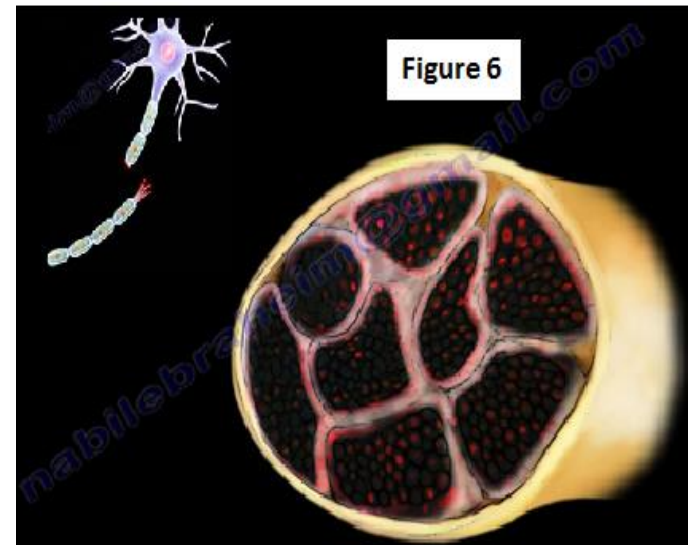
- Axon damage while the surrounding CT remains intact
- Occurs in crush injuries or displaced bone fractures causing disruption of the nerve cell axon
- Exhibits prolonged weakness with gradual return of function after **several months**
- There is partial or complete recovery the nerve



Classification

❑ Neurotmesis

- Occurs with knife lacerations, gunshot wounds or severe ischemic injuries
- Axon and surrounding CT are damaged
- The most serious nerve injury
- **No recovery** will take place
- Surgery is required



**TABLE
5-1****Classification of Nerve Injuries and Associated Findings**

Injury Type	Histologic Findings	Response	Recovery Pattern	Recovery Rate	Treatment
First degree (Seddon neuropraxia)	Transient ischemia, anoxia, $\pm$ segmental demyelination, intrafascicular edema	Conduction block	Complete	Fast (hours to weeks)	None indicated
Second degree (Seddon axonotmesis)	Axon and myelin interruption (intact endoneurium, perineurium, & epineurium)	Wallerian degeneration distal to injury	Complete	Slow (weeks)	None indicated
Third degree	Injury involves endoneurium (intact perineurium & epineurium)	Wallerian degeneration and some degree of neuron cell death	Variable	Slow (weeks to months)	Nerve exploration may be considered
Fourth degree	Injury involves endoneurium and perineurium	Wallerian degeneration, neuron cell death, neuroma formation and intrafascicular fibrosis	None	Spontaneous recovery not likely	Microneurosurgery
Fifth degree (Seddon neurotmesis)	Complete nerve transection, continuity disruption	Wallerian degeneration, cell death neuroma, fibrosis	None	No spontaneous recovery possible	Microneurosurgery

Treatment

- ❑ Open injuries of the facial nerve should be repaired surgically
 - Immediate primary repair as soon as possible
 - Delayed primary repair or early secondary repair is acceptable, when delay is necessary for certain reasons
- ✓ ***12 months is often considered as the limit for successful facial nerve repair***

Treatment

- Neurorrhaphy Procedures
 - Direct nerve repair
 - Cable nerve grafting
 - Nerve substitution techniques

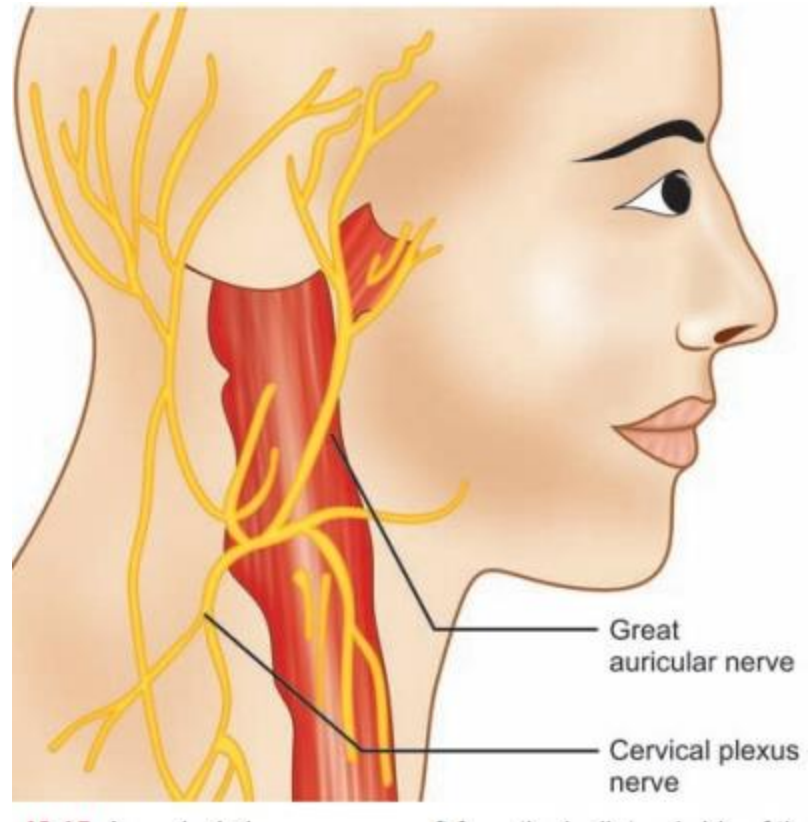
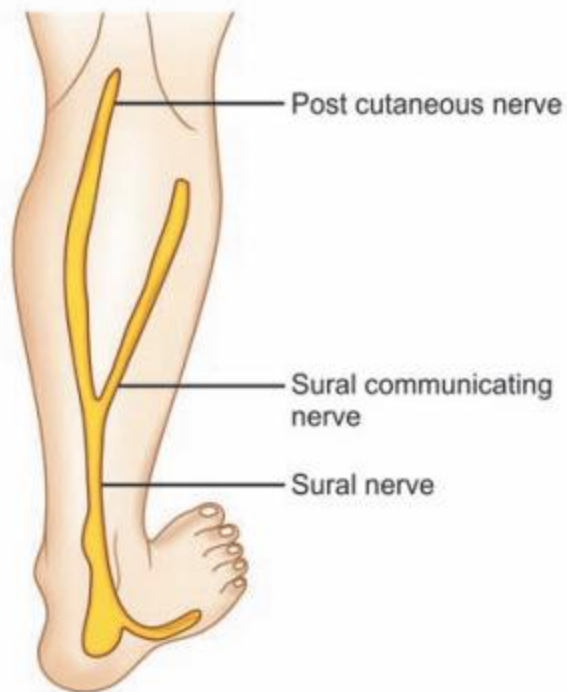
Neurorrhaphy

- ❑ Primary facial nerve repair
 - Direct end to end approximation
 - Indication
 - ❑ Surgical repair of the transected facial nerve
 - ❑ Indicated whenever the nerve length is sufficient without undue tension <17mm loss.

Neurorrhaphy

- ❑ Autogenous nerve grafting is done
 - If there is a gap between the proximal and distal part of the nerve with some part of the nerve loss
 - Hypoglossal nerve
 - Sural nerve
 - Branches of cervical plexus
 - A great auricular nerve

Donor nerves

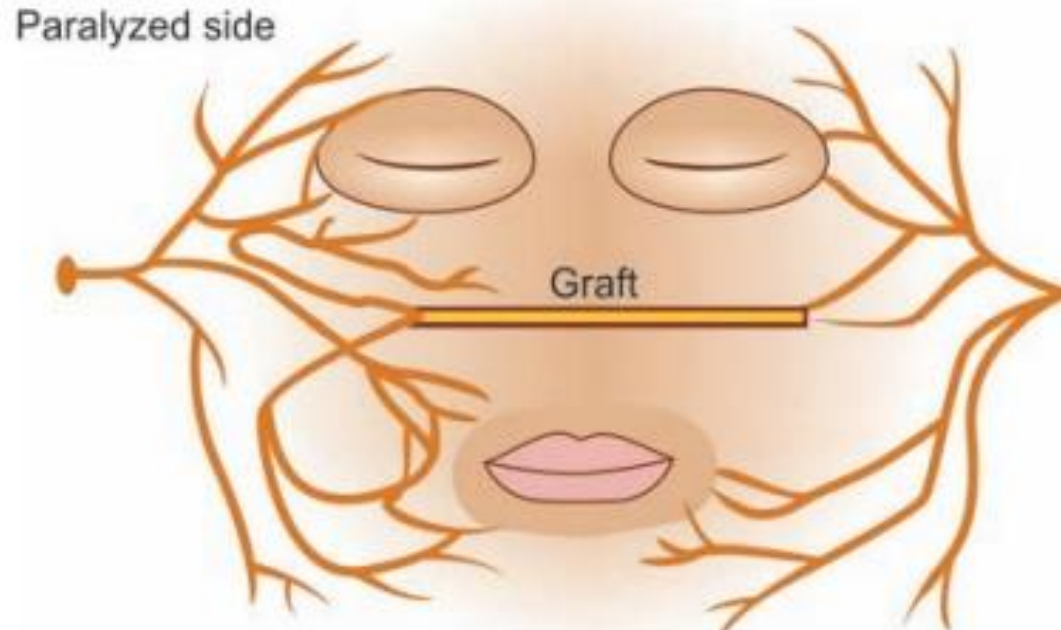


Neurorrhaphy

□ *Reanimation technique*

- Reanimation or cross face nerve grafting technique
- has been effective in restoring some facial muscle function
 - Hypoglossal - facial anastomosis
 - Cross - face anastomosis
 - surgical redirection of the accessory nerve into facial nerve

Reanimation technique



On the non paralyzed side intact buccal branch can be connected to the paralyzed stem of the facial nerve with a sural nerve graft

Complications

- ❑ Hematoma
- ❑ Infection
- ❑ Donor site morbidity
 - Hypoglossal – tongue weakness
 - Great auricular harvesting – ear numbness
 - Sural nerve harvesting – lateral leg numbness

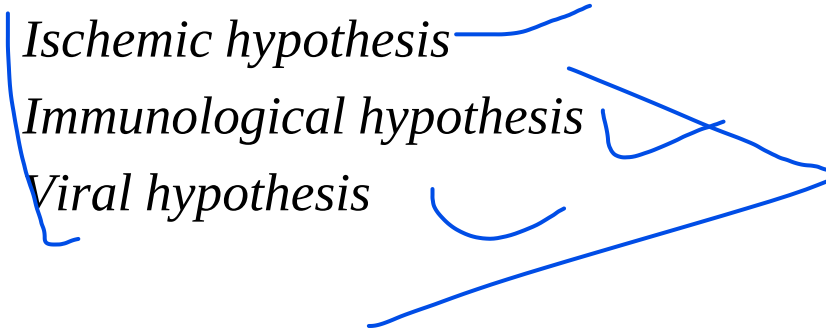
Facial palsy

- ❑ **Bell' s palsy**
- ❑ **Definition ?**
- ❑ An idiopathic paresis or paralysis of the facial nerve of sudden onset .

Epidemiology

- ❑ Sex : W > M
- ❑ 3.3 times more common in pregnant women
- ❑ Age: at any age more common in middle age
- ❑ 31% pts are DM
- ❑ Side involvement – equal/Unilateral

Etiology

- ❑ Although Bell's palsy by definition is idiopathic condition
 - ❑ Five hypotheses have been proposed regarding etiology
 - *Rheumatic hypothesis*
 - *Cold hypothesis*
 - *Ischemic hypothesis*
 - *Immunological hypothesis*
 - *Viral hypothesis*
- 
- ✓ Many times a combination of factors may play a role

Hypothesis

- *Rheumatic hypothesis* - Berard (1936)
 - Proposed that rheumatic swelling may press the nerve against the walls of the fallopian canal (Obsolete theory)

- *Cold hypothesis* - Proposed by Charles Bell
 - Exposure to extreme cold or cold draught

Hypothesis

□ *Ischemic hypothesis*

- disturbed circulation in the vasoneurosum causes facial paralysis

□ *Immunological hypothesis*

- sensitization of lymphocytes to peripheral nerve myelin gives rise to a cell mediated autoimmune response
- It can result in autoimmune demyelization of cranial neuritis

Hypothesis

□ *Viral hypothesis* –

- It is believed that Bell's palsy is caused by subclinical herpes zoster or herpes simplex infection
- The combination of **facial paralysis** and **herpes** is known as the *Ramsay Hunt syndrome*
- X-zed by Herpetic eruptions
 - skin, ear, Tm,
 - palate, fauces,
 - soft palate and tongue

Clinical features

- ❑ There is sudden onset, after awakening early in the morning
- ❑ Onset is usually as pain in the region of stylomastoid foramen
- ❑ Development of unilateral facial paralysis
- ❑ Inability to close eye on the affected side Bells sign
- ❑ Mouth is drawn to the opposite side
- ❑ Dribbling of the saliva & fluids from the angle of mouth

Clinical features

- ❑ Impairment of sensation
- ❑ Whistling is impossible
- ❑ Speech becomes slurred
- ❑ Obliteration of the nasolabial fold
- ❑ Occasionally there is loss or alteration of taste

Clinical features



Management

- Treatments includes
 - Corticosteroid therapy
 - Antiviral agents
 - eye care
 - Physiotherapy

Treatment

- ❑ Corticosteroid therapy
 - Corticosteroids regimens should be initiated with 72 hours of symptoms
 - Prednisolone 1mg/kg po 10 – 14 days with tapering dose
- ❑ Dosage for a 60 kg person— prednisolone 10 mg tablets—3 tablets twice daily for 4 days, then 2 tablets twice daily for 4 days, then 1 tab twice daily for 4 days, then 1 tablet daily for 4 days and stop.
- ❑ Vitamins B1,B6,B12 may be administered.

Treatment

- ❑ Antiviral agents regimen
 - HSV- Acyclovir 400mg poX5/daily for 10 days or valacyclovir 500mg po BID for 05 days
 - VZV- Acyclovir 800mg poX5/daily for 10 days or valacyclovir 1000mg po TID for 5 days

Management

□ Physiotherapy

- indicated to maintain the muscle tone
- should be instituted as early as possible
- It consists of electrical stimuli by galvanism, gentle massage & facial exercises

Treatment

- ❑ Eye care - if incomplete eye closure is present
 - Artificial lubrication with artificial tear
 - Eye patch
 - taping the eye or
 - tarsorrhaphy to prevent visual loss from exposure keratitis

Prognosis

□ Unilateral facial weakness

- Gradually worsen 2 to 3 days ,reaching maximum in about 2 weeks
- Remission begins within three weeks of onset in 85 percent of cases
- Spontaneous recovery is known to occur in Bell's palsy?
- Without treatment...?
- DDX should be made in the facial paralysis due to other etiological factors
- Prognosis in RHS....50%

Complications

- ❑ Possible sequelae include:-
 - ✓ Ageusia
 - ✓ Facial spasm or facial pain
 - ✓ Corneal infection/abrasion.
 - ✓ Synkinesis-an involuntary muscular mov't occurring simultaneously with voluntary mov'ts.

Reading

- ❑ Supraneuclear facial nerve palsy
- ❑ Infranuclear facial nerve palsy
- ❑ Why?
- ❑ MRS
- ❑ Crocodile tearing syndrome

Reference

- ❑ Nelima Anil Malik. Textbook of Oral and Maxillofacial Surgery 3rd edition
- ❑ *Raymond J. Fonseca. Oral and Maxillofacial Surgery 3rd Edition*

Thank u!